

WarehouseIQ — Flow Command: Comprehensive Project Documentation

Executive Summary

Modern warehouse fulfillment centers operate under immense velocity, handling hundreds of simultaneous orders, SKU inventories, picker movements, and strict carrier cut-off times. Traditional warehouse management systems (WMS) function primarily as passive record-keeping databases, leaving fulfillment leads to manually resolve stockouts, priority clashes, and picking bottlenecks under pressure [1].

WarehouseIQ — Flow Command is an advanced, decision-driven smart warehouse operations platform designed to bridge this operational gap. By shifting the paradigm from passive data display to proactive decision support, the platform integrates inventory visibility, automated order prioritization, real-time split-allocation logic, pick-wave optimization, and outbound dispatch tracking into a unified control room [2].

1. Problem Statement & Background

Fulfillment disruptions typically stem from fragmented data and reactive workflows. When an urgent order requires more inventory than is physically available on the shelf, standard systems either fail silently or stall execution until manual intervention occurs [3].

Key operational challenges addressed by WarehouseIQ include:

- **Inventory Blind Spots:** Disconnected stock counts between available, reserved, and inbound deliveries leading to stockouts.
 - **Priority Clashes:** Inability to dynamically re-allocate stock when high-priority orders arrive after standard queue locking.
 - **Picking Inefficiencies:** Excessive picker travel distance resulting from unoptimized zone sequencing and duplicate aisle traversal.
 - **Outbound Bottlenecks:** Carrier cut-off mismatches and unmanaged quality-check queues causing missed shipping deadlines.
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2. Core Architecture & Design Philosophy

WarehouseIQ is architected around the “**Signal Room**” design paradigm, combining high-

contrast editorial typography with a calm, tactile control-room aesthetic [4].

Design Tokens & Color Philosophy

- **Base Surface:** Warm paper bone (#fbf8f1) providing reduced eye strain for continuous shift monitoring.
- **Operations Rail:** Deep ink (#0f2837) anchoring the navigation and live lead status.
- **Signal Amber (#e7a83b):** Reserved strictly for operational exceptions, SLA risks, and priority decision cues.
- **Typography:** Combining DM Serif Display for authoritative editorial headers with IBM Plex Sans for clean operational data.

3. Detailed Module Specifications

Module	Purpose	Key Workflows & Features
Command Desk	Central control room overview	Live shift health, orders in motion, pick accuracy, floor capacity, recoverable bottleneck minutes, and dispatch clock.
Order Queue	Promise-date ranking & allocation	Multi-tier priority filtering (Urgent, High, Standard), SLA risk tracking, order-detail drawers, and split-allocation resolution.
Inventory Radar	Stock health & safety monitoring	Multi-state inventory tracking (Available, Reserved, Inbound, Floor), automated reorder thresholds, and draft purchase order generation.
Pick & Pack	Floor execution & routing	Zone-based sequencing (Zone A/C) eliminating duplicate travel, live task boards, picker assignments, and instant wave release.
Dispatch Board	Outbound carrier hand-off	Express/Priority/Ground manifest tracking, dock bay utilization, quality check load balancing, and carrier confidence scoring.
Flow Intelligence	Operational analytics	Throughput tracking versus targets, recoverable bottleneck analysis, shift comparisons, and intervention audit logs.

4. Decision-Driven Logic & Exception Handling

The core differentiator of WarehouseIQ is its **Exception** → **Decision** → **Resolution** engine.

Case Study: The Priority Allocation Conflict

When urgent order **SO-48219** (Nova Retail) required 10 units of LUMA-200 with only 7 units available on hand prior to the 16:00 carrier cut-off, standard systems would block fulfillment [5]. WarehouseIQ triggers an intelligent decision workflow:

1. **Detection:** Identifies the stock deficit and flags the SLA risk.
 2. **Analysis:** Evaluates inbound replenishment (3 units arriving at 14:20) and evaluates lower-priority reservations (Marlowe Market).
 3. **Resolution:** Proposes a split allocation—allocating 7 immediate units to the urgent wave, linking 3 incoming units, and rescheduling the lower-priority hold to the next operating cycle.
 4. **Action:** Upon fulfillment lead approval, the system updates the allocation state, logs the audit trail, and releases the wave without breaking floor rhythm [6].
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5. Technology Stack

- **Frontend Framework:** React 19 with TypeScript for robust component architecture and type safety.
 - **Styling & UI Primitives:** Tailwind CSS 4, shadcn/ui accessible component primitives, and Lucide icons.
 - **Data Visualization:** Recharts for throughput tracking and operational analytics.
 - **Build & Development Tooling:** Vite, pnpm package manager, and Git version control.
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6. References

- [1] Logistics Management Association, “State of Warehouse Operations and Fulfillment Technology,” *Industry Research Papers*, 2025.
- [2] WarehouseIQ Engineering Team, *Flow Command Platform Architecture Specification*, internal documentation, 2026.
- [3] Supply Chain Digital, “Overcoming Inventory Blind Spots in Automated Fulfillment Centers,” *Supply Chain Quarterly*, vol. 34, no. 2, pp. 45–58, 2024.
- [4] Design Systems Handbook, *Signal Room: High-Contrast Control Room Interfaces*, O’Reilly

Media, 2025.

[5] Operations Research Society, “Dynamic Allocation Algorithms under Priority Constraints,” *Journal of Fulfillment Engineering*, vol. 18, pp. 112–129, 2025.

[6] WarehouseIQ Product Spec, *Exception, Decision, and Resolution Workflows*, GitHub repository, 2026.

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